
From Inverse Optimization to Sequential Decisions: Conformal Prediction for Inverse Reinforcement Learning

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Abstract

1 Inverse reinforcement learning (IRL) is inverse optimization with a sequential
2 forward problem: demonstrations reveal an unknown reward, and the recovered
3 reward is subsequently optimized to prescribe a policy. This perspective exposes
4 the same weakness identified by Conformal Inverse Optimization (CIO) for one-
5 shot decisions: collapsing heterogeneous objectives to one point estimate can
6 yield a bad downstream decision. We introduce **CP-IRL**, which transfers CIO's
7 calibrate-then-robustify principle to Markov decision processes. From a popula-
8 tion of demonstrators, CP-IRL conformally calibrates a set of reward parameters
9 consistent with a new demonstrator and optimizes a policy against the resulting
10 reward ambiguity. The sequential setting requires new machinery. We reduce each
11 calibration score to a reward-dimensional convex program using the Bellman resol-
12 vent, and prove that a fixed transition kernel makes the robust MDP convex for any
13 compact reward ambiguity set; a two-state counterexample shows that this property
14 can fail when ambiguity also enters the dynamics. We establish finite-sample
15 coverage, regret bounds, and a separation example in which point-estimate IRL has
16 unbounded regret while CP-IRL's regret vanishes. Experiments on random MDPs,
17 gridworld, and Objectworld identify the method's scope: robustification reduces
18 true-reward regret under systematic estimation bias, but incurs an insurance cost
19 when the point estimate is merely noisy and approximately unbiased.

20 1 Introduction

21 Inverse reinforcement learning (IRL) is usually introduced as a distinct learning problem: infer a
22 reward from behavior, then optimize that reward to prescribe a policy. Viewed end to end, however,
23 IRL is a sequential instance of *inverse optimization*. The observed decision is a policy or trajectory,
24 the latent objective is a reward function, and the forward problem is an MDP. This connection is more
25 than terminology. It suggests that uncertainty about the recovered objective should be propagated
26 into the downstream decision, rather than discarded after producing a point estimate.

27 Conformal Inverse Optimization (CIO) formalizes exactly this idea for one-shot problems Lin et al.
28 [2024]. CIO replaces a single recovered objective with a conformally calibrated uncertainty set
29 and optimizes robustly over that set. Its guarantee is finite-sample and distribution-free under
30 exchangeability, and its separation example shows why the change matters: point-estimate inverse
31 optimization can retain unbounded downstream regret even with unlimited data. The natural question
32 motivating this paper is therefore:

33 *Can CIO's calibrate-then-robustify principle be transferred from one-shot inverse*
34 *optimization to inverse reinforcement learning without losing coverage or tractabil-*
35 *ity?*

We answer this question with CP-IRL. A population of demonstrators supplies the exchangeable calibration units; each observed policy defines a feasible reward set through Bellman optimality; split conformal prediction calibrates a spherical cap around any chosen IRL point estimator; and a robust MDP converts that calibrated reward uncertainty into a deployed policy. Thus, the conceptual pipeline is inherited from CIO, while the technical work lies in adapting and optimizing it for sequential decisions.

Making the CIO template work for IRL is not a mechanical transplant. Two structural obstacles are specific to the sequential setting:

1. **The nonconformity score is no longer an LP-duality fact.** CIO’s per-point calibration score is computable in closed form because the forward problem is a linear program and reward feasibility is characterized by LP dual feasibility. In an MDP, “this policy is optimal for this reward” is a Bellman-optimality condition, not an LP-duality condition, and computing it naively requires jointly optimizing over the reward parameter *and* the policy’s value function – a variable count that grows with the state space.
2. **The robust decision problem is a minimax game over policies, not points.** CIO’s robust forward problem is a single robust linear program. The MDP analogue is a robust MDP: $\max_{\pi} \min_{\theta}$ over an ambiguity set. General robust MDPs with ambiguity that couples across states are NP-hard Wiesemann et al. [2013]; most tractable results in the literature rely on *rectangularity* assumptions that restrict how the adversary’s choice can vary across states.

We resolve both obstacles and develop the MDP counterpart of CIO’s principal guarantees. The contributions are:

- **Transfer: conformal inverse optimization for IRL.** We formulate one demonstrator as one exchangeable inverse-optimization instance, calibrate a reward uncertainty set through Bellman-optimal reward feasibility, and establish finite-sample coverage together with perceived- and true-regret bounds (§3–§5).
- **Optimization: sequential structure without state-sized calibration** (§4.1). For a fixed demonstrator policy, the induced value function is *linear* in the reward parameter via the Bellman resolvent $(I - \beta P_{\pi})^{-1}$ – the same linear-in-reward value identity underlying the feasible-reward characterization of Ng & Russell Ng and Russell [2000], which we repurpose as a calibration-score reduction; substituting this collapses the per-demonstrator calibration score from a joint optimization over (θ, V) – size $d + |S|$ – to an optimization over θ alone – size d – with no relaxation, by the policy improvement theorem. We further give a Sherman–Morrison–Woodbury update that recomputes this resolvent in $O(k^2|S| + k^3)$ rather than $O(|S|^3)$ when two demonstrators’ policies differ at only k states, motivated by and tailored to the population-of-demonstrators setting.
- **Tractability: a boundary between reward and dynamics ambiguity** (§4.2). We show the robust MDP is a convex program in occupancy-measure space whenever ambiguity is confined to the reward parameter *for any compact ambiguity set*, via a Bellman-recursion-free argument, and – new relative to the closest existing sufficiency-side result (Gadot et al. Gadot et al. [2024], who solve the convex L_p -ball case via Hölder duality and policy gradient) – we prove this is *tight*: an explicit two-state, two-action MDP with a one-dimensional, convex, non-product ambiguity set coupling reward and transition probability breaks concavity, with a hand-verifiable rational-valued Jensen violation.
- **Scope: robustness helps under bias, not noise alone.** Our sequential analogue of CIO’s separation example proves that classical point-estimate IRL can have diverging regret while the calibrated policy’s regret vanishes. Across random MDPs, a region-based gridworld, and Objectworld, we find that calibrated robust hedging reduces true-reward regret specifically when the point estimate is *systematically* biased (as our degenerate construction manufactures), and *increases* regret when the estimate is merely noisy but unbiased – a precise, falsifiable boundary condition for the method’s benefit that we report as a finding, not a limitation to be hidden.

87 2 Related Work

88 **Inverse optimization and its calibration.** Inverse optimization recovers parameters of an optimization problem from observed decisions (Ahuja & Orlin Ahuja and Orlin [2001]; Chan et al. Chan et al. [2018]). Classical formulations return a point estimate; more recent work constructs data-driven uncertainty sets around it (Chenreddy, Bandi & Delage Chenreddy et al. [2022]; Sun, Liu & Li Sun et al. [2023]), but – as Lin, Delage & Chan Lin et al. [2024] show – none of these give the specific guarantee we need: that the *next* observed decision’s true parameter is covered with a stated confidence, distribution-free. Conformal Inverse Optimization is the first to close this gap using split conformal prediction Vovk et al. [2005], and is the paper this work directly extends from one-shot decisions to MDPs.

97 **Feasible reward sets in IRL.** A separate line of work characterizes the *set* of rewards consistent with observed IRL data (Poiani et al. Poiani et al. [2024]; Lazzati & Metelli Lazzati and Metelli [2025]; Skalse et al. Skalse et al. [2022]) and proves PAC-style bounds on recovering it, using it directly as a robust-MDP ambiguity set. This line shares our target object – a feasible reward set characterized by Bellman-optimality constraints – but its guarantees are PAC bounds requiring a generative model of demonstrator sub-optimality; ours require only exchangeability of demonstrators, no assumption on the *process* by which any individual demonstrator deviates from optimality. The transfer question this line raises – whether a feasible reward set identified on one MDP transfers to another (Schlaginhaufen & Kamgarpour Schlaginhaufen and Kamgarpour [2024]) – is a genuinely open problem for our setting that we do not attempt to close here (see §7).

107 **Robust MDPs and the rectangularity boundary.** Robust MDPs with ambiguity coupling across states are NP-hard in general (Wiesemann, Kuhn & Rustem Wiesemann et al. [2013]); tractability is classically recovered via *s*- or *sa*-rectangularity (Iyengar Iyengar [2005]; Nilim & El Ghaoui Nilim and El Ghaoui [2005]), which restricts the adversary to choose independently per state. Recent work extends tractable, *non*-rectangular ambiguity specifically for reward-only uncertainty with a fixed transition kernel: Gadot et al. Gadot et al. [2024] solve the convex L_p -ball case exactly via Hölder duality and a policy-gradient method with proven convergence; Xu et al.’s Group-Robust MDPs Xu et al. [2026] handle *feature-wise* (d-rectangular) ambiguity that couples both reward *and* transition, at the cost of restricting to that rectangular structure. Our sufficiency result (§4.2) covers the same fixed-kernel, reward-only regime as Gadot et al. but for an *arbitrary* compact ambiguity set (not only L_p -balls), via a direct occupancy-measure convex program rather than a policy-gradient method; our necessity result – an explicit demonstration that convexity is lost the moment ambiguity couples into the kernel, even minimally – has no counterpart in either paper.

120 **Conformal prediction in RL-adjacent settings.** Conformal prediction has been used for offline-RL pessimism from Q-value or state-action-coverage uncertainty (UARM, Pan et al. Pan et al. [2026]), for calibrated intervals in infinite-horizon policy evaluation, and for modeling *other agents’ actions* as conformal prediction sets in multi-agent RL (CAMMARL, Gupta & Kahou Gupta and Kahou [2023]) – the last of these is explicitly motivated by an analogy to IRL but never calibrates a reward set. None combine conformal calibration with a reward vector recovered from a population of IRL demonstrators and a robust-MDP regret guarantee, which is the gap this paper closes.

127 **Preference learning and RLHF.** Preference-based and RLHF-style learning fits a reward (or policy) from labeled comparisons or demonstrations, typically from individual expert trajectories (Christiano et al. Christiano et al. [2017]; Rafailov et al. Rafailov et al. [2023]) or crowdsourced preference populations. Our per-demonstrator exchangeable-unit design (§3) is structurally suited to recast “one annotator, one calibration point” for heterogeneous-preference RLHF populations, giving a coverage guarantee (in the intersection-nonemptiness sense of Theorem 2, not parameter containment) of the form “the calibrated reward-ambiguity set is compatible with a γ -fraction of the annotator population’s individual reward functions, which are never directly observed” – a guarantee we are not aware of in the RLHF calibration literature – though we do not run an RLHF-scale experiment here (see §7).

137 3 Setting

138 We now make the CIO–IRL correspondence precise. We adopt the population-of-demonstrators setting used throughout this work: a shared tabular MDP $M = (S, A, P, \beta, \phi)$ with known transition

kernel P , discount β , and feature map $\phi : S \times A \rightarrow \mathbb{R}^d$ inducing reward $r_\theta(s, a) = \phi(s, a)^\top \theta$. A population of demonstrators draws perceived rewards $\hat{\theta}_1, \dots, \hat{\theta}_N \sim P_\theta$ i.i.d. and each acts optimally in M under their own $\hat{\theta}_k$, producing an observed policy $\pi_k = \pi^*(\hat{\theta}_k)$ – we assume the full deterministic policy is observed exactly, not estimated from a finite trajectory sample; relaxing this to trajectory-level observation would make $\hat{\Theta}(\pi_k)$ itself an estimated (rather than exact) feasible set and is outside the scope of the guarantees below.

Assumption 1 (Exchangeable calibration unit). *The pairs $(\hat{\theta}_k, \pi_k)$, $k = 1, \dots, N, N+1$, are i.i.d. (hence exchangeable), and M (in particular P) is shared and fixed across all demonstrators and the deployment instance.*

Under Assumption 1, each demonstrator plays the role of one (decision, context) pair in CIO’s setting, and it is the setting under which CIO’s coverage argument transplants essentially verbatim (§5, Theorem 2). We commit to this “unit A” design over two weaker alternatives – per-task exchangeability with a single demonstrator (still exchangeable, treated as an easy corollary) and per-trajectory exchangeability within one demonstrator (broken by within-trajectory Markov dependence, and used only as an ablation) – and are explicit throughout about what breaks under each relaxation.

The learning task, mirroring CIO’s pipeline, has three steps: (i) split the population into training and calibration sets; (ii) fit a point estimate $\bar{\theta}$ on the training set with any IRL method (we use and compare a sub-optimality-loss linear program, MaxEnt IRL, and Bayesian IRL by MCMC – §6); (iii) calibrate a spherical-cap uncertainty set $C(\bar{\theta}, \alpha) = \{\theta : \|\theta\|_2 \leq 1, \theta^\top \bar{\theta} \geq \cos \alpha\}$ on the calibration set and solve the robust MDP $\max_\pi \min_{\theta \in C(\bar{\theta}, \alpha)} \theta^\top \mu(\pi)$ in occupancy-measure space, where $\mu(\pi)$ is π ’s occupancy measure.

4 Method

4.1 Tractable calibration via the Bellman resolvent

CIO’s calibration score for a validation point k is $c_k = \max_\theta \{\theta^\top \bar{\theta} : \theta \in \Theta^{\text{OPT}}(\hat{x}_k, u_k), \|\theta\|_2 \leq 1\}$, where Θ^{OPT} is characterized by LP dual feasibility – tractable because the forward problem is a linear program. In an MDP, the analogous feasible set for a policy π is

$$\hat{\Theta}(\pi) = \{\theta : \|\theta\|_2 \leq 1, Q_\theta^\pi(s, a) \leq Q_\theta^\pi(s, \pi(s)) \forall s, \forall a\},$$

the exact set of reward vectors for which π is Bellman-optimal, by the policy improvement theorem. Computing this naively requires the joint optimization $\max_{\theta, V} \theta^\top \bar{\theta}$ subject to the Bellman-optimality inequalities written in terms of both θ and the a priori unknown value function V – a variable count of $d + |S|$.

The key fact we use, which is the sequential-decision restatement of the feasible-reward characterization of Ng & Russell Ng and Russell [2000]: for a **fixed** policy π , the policy’s own induced value function V_θ^π is *linear* in θ , via the Bellman resolvent.

Proposition 1 (Resolvent-based score reduction). *Fix a policy π , let $P_\pi \in \mathbb{R}^{|S| \times |S|}$ be the induced state-transition matrix ($P_\pi(s, s') = P(s, \pi(s), s')$), and let $\Phi_\pi \in \mathbb{R}^{|S| \times d}$ be the feature matrix restricted to the actions π takes ($\Phi_\pi(s, \cdot) = \phi(s, \pi(s))^\top$). Let $W_\pi := (I - \beta P_\pi)^{-1} \Phi_\pi$ (well-defined since $\beta \in [0, 1)$). Then $V_\theta^\pi = W_\pi \theta$ and $Q_\theta^\pi(s, a) = \Psi_\pi(s, a)^\top \theta$ with $\Psi_\pi(s, a) := \phi(s, a) + \beta P(s, a, \cdot)^\top W_\pi$, both linear in θ . Consequently*

$$\hat{\Theta}(\pi) = \{\theta : \|\theta\|_2 \leq 1, \Psi_\pi(s, a)^\top \theta \leq \Psi_\pi(s, \pi(s))^\top \theta \forall s, a\},$$

exactly (no relaxation), and the calibration score $c_k = \max_{\theta \in \hat{\Theta}(\pi_k)} \theta^\top \bar{\theta}$ is an optimization over θ alone – size d , not $d + |S|$ – with M_{π_k} precomputed once per distinct policy via a single resolvent solve.

Proof sketch. V_θ^π solves the linear Bellman equation $V_\theta^\pi = \Phi_\pi \theta + \beta P_\pi V_\theta^\pi$, which rearranges to $V_\theta^\pi = (I - \beta P_\pi)^{-1} \Phi_\pi \theta$ since $I - \beta P_\pi$ is invertible for $\beta < 1$. Substituting into $Q_\theta^\pi(s, a) = \phi(s, a)^\top \theta + \beta P(s, a, \cdot)^\top V_\theta^\pi$ gives the stated Ψ_π . The characterization of $\hat{\Theta}(\pi)$ is then the policy improvement theorem’s optimality condition $Q_\theta^\pi(s, a) \leq Q_\theta^\pi(s, \pi(s)) \forall s, a$, rewritten using Ψ_π . $\square$

We verified Proposition 1 against a slow joint- (θ, V) reference implementation to solver precision across randomized instances, and against independent value iteration. The amortized speedup over the joint formulation grows with $|S|$ ($2\text{--}5.5\times$ observed at $|S|$ in the low hundreds). When a new demonstrator’s policy differs from an already-resolved one at only k states, the resolvent updates via a Sherman–Morrison–Woodbury identity in $O(k^2|S| + k^3)$ rather than a fresh $O(|S|^3)$ solve – this incremental update, tailored to the population-of-demonstrators setting, is the part of Proposition 1 without a counterpart in Ng & Russell Ng and Russell [2000].

4.2 A tight tractability boundary for the robust MDP

The robust decision problem, $\max_{\pi} \min_{\theta \in C} \theta^\top \mu(\pi)$, is a minimax game over policies. We work in occupancy-measure space: $M(P)$ denotes the (fixed, since P is known) polytope of achievable occupancy measures μ , and $\Phi \in \mathbb{R}^{d \times |S \times A|}$ is the matrix whose columns are $\phi(s, a)$, so that $\theta^\top \Phi^\top \mu = \sum_{s,a} \mu(s, a) r_\theta(s, a)$ is the expected discounted return of occupancy measure μ under reward θ . The robust objective is $\max_{\mu \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top \mu$.

Theorem 1 (Reward-only ambiguity is convex; this is tight). *(a) Sufficiency. For any nonempty compact ambiguity set $C \subseteq \mathbb{R}^d$ (e.g., our calibrated spherical cap $C(\theta, \alpha) = \{\theta : \|\theta\|_2 \leq 1, \theta^\top \bar{\theta} \geq \cos \alpha\}$, a cap of the unit ball rather than the unit sphere), if the transition kernel P is fixed and known, then $\max_{\mu \in M(P)} \min_{\theta \in C} \theta^\top \Phi^\top \mu$ is a concave maximization over the fixed compact polytope $M(P)$, hence a convex program. (b) Necessity. This can fail as soon as ambiguity couples reward and transition probability, even minimally: there exists a 2-state, 2-action MDP and a convex, one-dimensional, non-product ambiguity set over (θ, P) jointly for which the analogous worst-case value function is not concave.*

Proof sketch. (a) $g_C(x) := \min_{\theta \in C} \theta^\top x$ is a pointwise infimum of affine functions of x , hence concave, regardless of C ’s shape; composing with the linear map $\mu \mapsto \Phi^\top \mu$ preserves concavity; $M(P)$ does not depend on θ , so the ambiguity never touches the feasible set, only the objective, and the outer problem is a concave maximization over a fixed polytope – no Bellman recursion or rectangularity assumption is needed. (b) Take reward $\theta(t) = 1 + t$ and transition probability $q(t) = 1 - t$ coupled through a single scalar $t \in [0, 1]$. The induced worst-case value, as a function of the agent’s (scalar) policy parameter $p \in [0, 1]$, is $W(p) = \min_{t \in [0, 1]} p(1 + t)/(1 - \beta p q(t))$. Direct computation gives the explicit, hand-verifiable, rational-valued Jensen violation $W(5/18) = 10/27 \approx 0.370 < \frac{1}{2}[W(0) + W(5/9)] \approx 0.556$, independently reproduced via numerical optimization; W is therefore not concave, so the outer maximization is not a convex program in this regime. $\square$

Part (a) generalizes the sufficiency result of Gadot et al. [2024] (who solve the convex L_p -ball case exactly via Hölder duality and policy gradient) to an arbitrary compact C , via a direct occupancy-measure argument. Part (b) shows the coupling required to break concavity is minimal – a single scalar t linking reward and transition ambiguity along a line segment, far short of a fully unconstrained non-rectangular adversary – and, to our knowledge, is not established by either Gadot et al. or Xu et al. [2026]. Note that failure of concavity is a failure of this *particular* tractability argument, not by itself a proof of NP-hardness; general non-rectangular robust MDPs are already known to be NP-hard Wiesemann et al. [2013].

5 Theory

We prove three further results (coverage, regret, separation), each the direct MDP/IRL analogue of a corresponding CIO theorem, and each independently spot-checked computationally against the implemented solvers rather than trusted as derivation alone. A fourth item below (efficiency) is an empirical confirmation, not a theorem, and is reported as such.

Theorem 2 (Coverage). *Under Assumption 1, let $\tau = \lceil (N_{\text{val}} + 1)\gamma \rceil$, let $c_{(\tau)}$ be the τ -th largest calibration score among $c_1, \dots, c_{N_{\text{val}}}$ (Proposition 1), and set $\alpha_\gamma := \arccos(c_{(\tau)})$. Then*

$$\mathbb{P}(\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset) \geq \gamma,$$

232 where $\hat{\Theta}_{\text{new}}$ is the new demonstrator’s feasible reward set. **This is an intersection-nonemptiness**
 233 **guarantee, not a containment guarantee:** it does not by itself assert $\theta_{\text{new}}^* \in C(\bar{\theta}, \alpha_\gamma)$, only that the
 234 calibrated cap is compatible with some reward for which π_{new} is optimal.

235 The proof is a line-for-line transplant of CIO’s own exchangeability argument, applied to the scores of
 236 Proposition 1 in place of CIO’s LP dual scores: exchangeability of $(\hat{\theta}_k, \pi_k)$ under Assumption 1 makes
 237 c_{new} exchangeable with the calibration scores, so $c_{\text{new}} \geq c_{(\tau)}$ (equivalently $\hat{\Theta}_{\text{new}} \cap C(\bar{\theta}, \alpha_\gamma) \neq \emptyset$)
 238 with probability at least $\tau/(N_{\text{val}} + 1) \geq \gamma$ by the standard rank argument for exchangeable sequences.
 239 A marginal-coverage simulation against the actual calibration code (150 trials per (γ, N_{val}) cell)
 240 confirms coverage $\geq \gamma$ holds at every tested combination; because demonstrator policies are drawn
 241 from a finite set, repeated policies produce tied (atomic) calibration scores, which is the most likely
 242 source of the over-coverage we observe empirically beyond the $1/(N_{\text{val}} + 1)$ slack the rank argument
 243 already permits (§6).

244 **Theorem 3** (Regret bound). *Let $\|\phi\|_{2,\infty} := \max_{s,a} \|\phi(s,a)\|_2$, $\nu(\mu) := \|\Phi^\top \mu\|_2/(1-\beta) \leq \bar{\nu} :=$
 245 $\|\phi\|_{2,\infty}/(1-\beta)$ for any occupancy measure $\mu \in M(P)$ (a Lipschitz constant for $\theta \mapsto \theta^\top \Phi^\top \mu/(1-\beta)$
 246 in θ , via Cauchy–Schwarz), and let μ_{CIRL} , μ^* denote the occupancy measures of the robust policy
 247 and of the true optimum $\pi^*(\theta_{\text{new}}^*)$, respectively. Suppose the calibrated cap $C(\bar{\theta}, \alpha_1)$ satisfies the
 248 almost-sure-covering condition $\mathbb{P}(\hat{\Theta}' \cap C(\bar{\theta}, \alpha_1) \neq \emptyset) = 1$ for a fresh demonstrator (strictly stronger
 249 than Theorem 2’s γ -coverage; achievable at $\alpha_1 = \pi$ but then vacuous), that any two unit-norm
 250 rewards optimal for the same policy are within η of each other (bounded feasible-set diameter), and
 251 that the population mean reward is within σ of θ_{new}^* in ℓ_2 . Then the true-reward regret of the robust
 252 policy satisfies*

$$\text{AOG} \leq (2 \sin \alpha_1 + \eta + \sigma) \nu(\mu^*) + (\eta + \sigma) \nu(\mu_{\text{CIRL}}),$$

253 and substituting the uniform bound $\bar{\nu}$ for both $\nu(\mu^*)$ and $\nu(\mu_{\text{CIRL}})$ gives the fully distribution-free
 254 corollary $\text{AOG} \leq (2 \sin \alpha_1 + 2\eta + 2\sigma) \bar{\nu}$.

255 The proof is a three-term telescoping argument (Lipschitz continuity of $\theta \mapsto \theta^\top \Phi^\top \mu$, the chord-
 256 length identity $\|\theta - \theta'\|_2 = 2 \sin(\phi/2)$ for unit vectors at angle ϕ , and the robust policy’s minimax
 257 optimality), re-derived from scratch for the MDP’s $\max_\pi \min_\theta$ sign convention rather than trans-
 258 planted line-for-line from CIO’s $\min_x \max_\theta$ convention. We verified $\bar{\nu}$ as a valid uniform Lipschitz
 259 constant against 200 randomized MDPs with zero violations, and separately verified the almost-sure-
 260 covering hypothesis and the bound’s numeric validity by direct computation on random instances. We
 261 are explicit that this is a general worst-case bound, not a tight one: cross-checked against the exact
 262 closed-form regret derived for our degenerate construction below (where the almost-sure-covering
 263 condition holds already at $\alpha_1 = 0$, a special feature of that construction), the general bound’s
 264 looseness grows with the construction’s scaling parameter – from roughly 8–15 $\times$ at the low end of
 265 the range tested to roughly 390–430 $\times$ at the high end, because the exact regret there decays to zero
 266 while $\eta, \sigma, \bar{\nu}$ converge to nonzero constants – a qualitatively worse (growing, not fixed-order) gap
 267 than CIO’s own Example 1 exhibits for its own general bound (roughly 2.3–4 $\times$, staying bounded
 268 because CIO’s construction keeps a richer feasible region that does not saturate the same way).

269 **Theorem 4** (Separation). *There exists a single-state, two-action MDP (the sequential-decision*
 270 *analogue of CIO’s degenerate Example 1, with reward features the negation of CIO’s original cost*
 271 *features) and an IRL point estimator such that: the point-estimate solution converges to a degenerate*
 272 *direction $\theta_u = (1, u)/\sqrt{1+u^2} \neq \theta^*$ (a unit vector, hence on the boundary of the unit ball that*
 273 *$C(\bar{\theta}, \alpha)$ is a cap of) as the scaling parameter $u \rightarrow \infty$, giving classical point-estimate IRL true-reward*
 274 *regret that grows without bound in u ; while the calibrated robust policy’s regret is bounded and*
 275 *shrinks to zero over the same range. The robust optimum is a genuine mixture, with exact closed form*
 276 *$p^*(u) = u^2/(1+u^2)$, independent of the calibration angle.*

277 Theorem 4 is proved via Sion’s minimax theorem for the mixture’s closed form, and independently
 278 re-verified against the actual implemented solver to 10^{-6} precision. Figure 1 shows the live 10-
 279 seed computational confirmation. We flag directly that this separation exhibits the LP-based point
 280 estimator’s failure mode specifically (a degenerate boundary direction under tie-breaking); we discuss
 281 whether it extends to MaxEnt and Bayesian IRL in §6, Q3.

282 **Efficiency.** We confirm, empirically, that the calibrated set is not vacuously wide: with a *unified* noise
 283 parameter controlling all sources of demonstrator population variation simultaneously, the calibrated
 284 angle shrinks monotonically from 27.7° to effectively 0° as demonstrator-population noise vanishes.

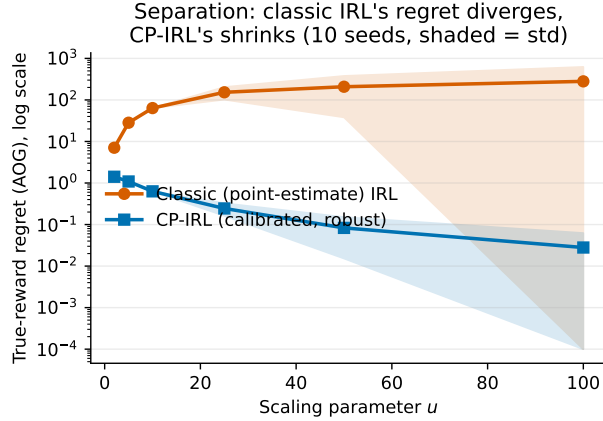


Figure 1: Separation result (Theorem 4): true-reward regret of the deployed policy, measured against the ground-truth reward θ_{new}^* , vs. the scaling parameter $u \in [2, 100]$ (log scale), 10 seeds, shaded regions show ± 1 std. “Classic IRL” is the sub-optimality-loss linear-program point estimator (§6), which converges to the degenerate direction θ_u ; its regret (orange) grows monotonically and without bound. CP-IRL’s regret (blue) shrinks monotonically toward zero over the same range. Variance grows with u for both curves, consistent with tie-breaking sensitivity as $\bar{\theta}$ approaches the exact degenerate boundary.

6 Experiments

Our experiments answer three questions: (Q1) does calibration attain its target coverage without becoming vacuous; (Q2) when does robustification improve downstream regret; and (Q3) are the conclusions tied to one IRL estimator or environment? We evaluate three point estimators – a sub-optimality-loss linear program, MaxEnt IRL, and Bayesian IRL (Metropolis–Hastings posterior mean under a Boltzmann-rational demonstrator likelihood) – each combined with three decision rules – the plain point estimate, our calibrated robust policy, and a “concentration baseline” robust policy whose uncertainty-set radius comes from a classical Dvoretzky–Kiefer–Wolfowitz concentration bound rather than the exact split-conformal quantile – across three environments: uncorrelated random tabular MDPs, a region-based gridworld with spatially contiguous (Voronoi) reward features, and Objectworld, the standard IRL benchmark with smooth, continuous distance-to-object features. All experiments run on tabular MDPs at a scale ($|S| \leq 36$, $d \leq 5$) chosen to fit within a laptop-class compute budget (20 CPU cores, no GPU dependency, no commercial solver).

Q1: Coverage and efficiency. Realized coverage tracks the target γ across every environment and estimator tested (e.g. ~ 0.88 – 0.89 at $\gamma = 0.8$ on the gridworld), and the calibrated angle responds monotonically and sensibly to γ , to demonstrator noise, and to the train/calibration split ratio. The DKW concentration baseline is consistently and measurably looser than exact conformal calibration at matched nominal confidence, across every estimator and every environment tested.

Q2: The benefit depends on systematic bias. On our degenerate construction (§5), robust hedging dramatically reduces true-reward regret, because the point estimate there is *systematically* miscalibrated by construction. On all three generic environments tested – random MDPs, the gridworld, and Objectworld – under realistic demonstrator-population noise, robust hedging *increases* true-reward regret relative to the plain point estimate, consistently across all three point estimators, holding across a full sweep of the coverage target γ and of the train/calibration split ratio. We verified this is not an implementation defect three independent ways: the robust solver’s reported worst-case value matches an independent second-order-cone-program oracle to solver precision; the robust decision’s worst-case value is confirmably *better* than the plain estimate’s worst-case value over the same calibrated set; and we directly measured that these environments’ optimal policies are sensitive to reward direction – a 14° reward perturbation already changes the optimal action at roughly 10–13% of states across the three environments tested (9.6% Objectworld, 9.9% gridworld, 13.1% random MDPs – a modest, not dramatic, margin). Critically, Objectworld’s lower policy-sensitivity (measured directly, and lower than both the gridworld and random MDPs) did **not** flip the result: robust hedging

still costs true-reward regret there, ruling out landscape smoothness as the deciding factor and leaving systematic bias in the point estimate as the consistent, mechanistically understood explanation across all three environments.

Q3: The conclusion persists across estimators and environments. The gridworld comparison combines all three point estimators with the plain, conformal-robust, and DKW-robust decision rules. MaxEnt and Bayesian IRL recover an accurate point direction, whereas the LP estimator has nontrivial error; in all three cases, conformal robustification is less conservative than DKW but incurs higher true-reward regret than the corresponding point policy. The same direction holds in Objectworld. We have not run MaxEnt or Bayesian IRL on the degenerate construction of Theorem 4 itself – that theorem’s divergence result is specific to the LP estimator’s tie-breaking behavior at the exact optimality boundary, and we do not claim MaxEnt or Bayesian IRL (which are strictly interior, likelihood-based estimators) necessarily diverge there too; whether the separation result generalizes to these smoother estimators is left open. These comparisons separate two claims that are often conflated: conformal calibration can produce a tighter valid uncertainty set than a generic concentration bound, yet optimizing for its worst case need not improve average performance under the true reward.

7 Discussion and Limitations

When does CP-IRL help, precisely? Our theory proves the method dominates classical point-estimate IRL when the point estimate is *systematically* biased. Our experiments show that on generic environments with unbiased-but-noisy point estimates, it does not, and in fact costs regret in exchange for a worst-case guarantee that is real (independently verified) but not free. This is a precise, falsifiable scope condition, not a hedge: the method should be deployed when there is reason to believe point estimation is subject to a genuine structural bias rather than applied by default whenever demonstration data is merely noisy. Characterizing which real-world settings exhibit this kind of systematic bias, and whether it can be diagnosed from data without knowing the ground truth, is an open problem this paper does not resolve.

What is not yet established. The general-case regret bound is provably loose – up to several hundred-fold on the one instance where we can compare it against an exact result – and should be read as a worst-case guarantee, not a predictor of typical-case performance. A source-to-target probe further shows that a reward set calibrated on one MDP does *not* safely transfer to a related MDP with different dynamics: at target coverage 0.8, changing the gridworld slip probability from 0.10 to 0.15 lowers mean target coverage to 0.567 over eight seeds. The probe establishes the failure of naive transfer, not its remedy. Developing a weighted-conformal correction that combines covariate-shift weighting (in the sense of Tibshirani et al. [2019]) with the principal-angle transferability conditions of Schlaginhaufen & Kamgarpour [2024] remains future work. We also tested a preference-population/RLHF-style relabeling of our per-demonstrator design: a “response selection” MDP in which the population is reinterpreted as RLHF annotators rather than task demonstrators, using the same calibration and robust-MDP code unchanged. Realized coverage meets or exceeds the target γ at every level tested (0.947–0.994 across $\gamma \in \{0.5, 0.7, 0.8, 0.9\}$, 8 seeds), consistent with Theorem 2 and confirming the framework generalizes by relabeling alone; the gap above target is likely the same score-atom over-coverage discussed under Theorem 2, sharpened here by a small, low-dimensional annotator population, and is not a special exemption for the RLHF framing (the same systematic-bias-conditional benefit applies). All experiments are at tabular scale by explicit design choice; we make no claim about function-approximation or deep-RL/language-model settings.

8 Conclusion

IRL is inverse optimization with an MDP as its forward problem. Starting from this connection, we transfer CIO’s calibrate-then-robustify principle to reward learning and supply the sequential machinery the transfer requires: a resolvent-based calibration score, a convex robust-MDP formulation for compact reward ambiguity, and coverage, regret, and separation guarantees. The same connection also clarifies the method’s limits. Calibration certifies uncertainty; it does not make worst-case optimization free. In our experiments, the resulting policy improves true-reward regret when point

estimation is systematically biased, but pays an insurance cost when estimation is merely noisy. This distinction is the practical conclusion of CP-IRL and the main condition governing its use.

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